

PHYS 217 E&M 1

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1 Vector Analysis

1.1 Vector Algebra

Definition 1.1. A **vector** is a quantity that has both direction and magnitude.

Definition 1.2. A **scalar** is a quantity that only has magnitude.

1.1.1 Vector Operations

There are four vector operations:

Addition:

- $\mathbf{A} + \mathbf{B}$ is the vector from the start of $\mathbf{A}$ to the end of $\mathbf{B}$
- Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- Subtracting a vector is just adding its opposite: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Multiplication by Scalar:

- $a\mathbf{A}$ is a vector with the same direction as $\mathbf{A}$ but a times the magnitude
- Distributive: $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$

Dot Product (Inner Product or Scalar Product):

- $\mathbf{A} \cdot \mathbf{B} \equiv AB \cos \theta$
- Results in scalar
- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A and the projection of $\mathbf{A}$ along $\mathbf{B}$

Cross Product of Two Vectors:

- $\mathbf{A} \times \mathbf{B} \equiv AB \sin \theta \hat{\mathbf{n}}$ where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. The direction
- is determined by the **right hand rule**, where you curl your right hand from $\mathbf{A}$ to $\mathbf{B}$ and take the direction of your thumb
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$
- Not commutative: $\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$
- Result is a **pseudovector** and behaves like a vector under rigid transformations like rotations, but not under certain nonrigid transformations. Notably, $-\mathbf{A} \times -\mathbf{B} = \mathbf{A} \times \mathbf{B}$
- Geometrically, $|\mathbf{A} \times \mathbf{B}|$ is the area of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$

When working in Cartesian coordinates, vectors can be expressed in terms of **basis vectors**:

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

with A_x , A_y , and A_z being the **components** of $\mathbf{A}$.

Vector operations can be done with the components, but the reader should already know how to do this.

There are two types of triple vector products:

Scalar Triple Product:

- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$
- Volume of parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$

Vector Triple Product:

- **BAC-CAB rule:** $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

1.1.2 Nomenclature

Definition 1.3. The **position vector** of a point is the vector to that point from the origin:

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

Definition 1.4. The **infinitesimal displacement vector** from (x, y, z) to $(x + dx, y + dy, z + dz)$ is:

$$d\mathbf{I} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$$

Definition 1.5. A **separation vector** is the vector from a **source point** to a **field point**:

$$\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'$$

1.1.3 Vector Transformations

Coordination transformation from rotation about an arbitrary axis can be represented by the use of a **rotation matrix**:

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

or more concisely:

$$\bar{A}_i = \sum_{j=1}^3 R_{ij} A_j$$

Definition 1.6. A transformation is **unitary** if it does not change the magnitude of the object on which it operates.

If rotation matrix $\overleftrightarrow{R}$ is unitary, then:

$$\sum_i^3 R_{ij} R_{ik} = \delta_{jk}$$

where:

$$\delta_{jk} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the Kronecker delta.

Definition 1.7. Parity is a property relating to $(\cdot)$. If a system stays the same after a parity transformation, it is said to have even parity. If it is inverted, it is said to have odd parity.

1.2 Differential Calculus